Gold Future/ETF Analysis

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2025

Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

Maturity	**Slope**	**Intercept**	R^2	**RMSE**
1-Month	0.9996	—	0.9494	—
2-Month	1.0024	—	0.9471	—
6-Month	1.0032	—	0.9407	—
12-Month	0.9478	—	0.8399	—

The slopes for the 1, 2, and 6-month contracts are all very close to 1, indicating a strong correlation with spot gold returns. The 12-month contract, however, has a noticeably lower slope of 0.9478 and a significantly lower R^2 of 0.8399. This may be a result of the lower liquidity and trading volume of the longer-dated contract, making it less sensitive to daily spot price movements.

Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

Maturity	**Slope**	**Intercept**	R^2	**RMSE**
1-Month	0.94314	2.4191×10^{-5}	0.80947	0.00530
2-Month	0.94301	1.0000×10^{-5}	0.80911	0.00530
6-Month	0.94348	3.1797×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.2333×10^{-5}	0.80984	0.00529

The slopes are also very close to 1, but consistently lower than the newer data. The R^2 values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity.

Table 3: New Data: Slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	**1-Month**	**2-Month**	**6-Month**	**12-Month**
Slope	1	0.9996	1.0024	1.0032	0.9478
	5	1.0074	1.0102	1.0106	0.9875
	10	1.0022	1.0040	1.0034	0.9897
	15	1.0031	1.0049	1.0033	0.9920
R^2	1	0.9494	0.9471	0.9407	0.8399
	5	0.9842	0.9840	0.9812	0.9567
	10	0.9915	0.9910	0.9878	0.9747
	15	0.9939	0.9933	0.9899	0.9807

The slopes and R^2 values generally increase with longer holding periods but decrease with longer-dated contracts.

Table 4: Leung & Ward (2015) Data: Slopes and R^2 from regressions of futures returns vs. gold returns over different holding periods.

	Days	**1-Month**	**2-Month**	**6-Month**	**12-Month**
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of $1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is $6.00 for the out of sample period.

Table 5: New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
1-m	0.00256	0.99743				$2.65	$5.15
2-m	0.00676		0.99324			$3.67	$5.75
12-m	0.04509				0.95491	$18.84	$17.95
1-m, 2-m	0.00168	1.21224	-0.21393			$2.61	$5.06
1-m, 6-m	0.00165	1.04753		-0.04918		$2.61	$4.88
1-m, 12-m	0.00189	1.01446			-0.01635	$2.66	$4.95
2-m, 6-m	0.00330		1.23075	-0.23405		$3.18	$4.63
2-m, 12-m	0.00310		1.09461		-0.09770	$3.26	$4.99
6-m, 12-m	0.00446			1.76090	-0.76537	$6.07	$12.15
1-m, 2-m, 6-m	0.00156	1.14088	-0.11333	-0.02911		$2.60	$4.94
1-m, 2-m, 12-m	0.00170	1.23840	-0.24373		0.00363	$2.64	$4.99
1-m, 6-m, 12-m	0.00197	1.10522		-0.17107	0.06388	$2.62	$4.54
2-m, 6-m, 12-m	0.00327		1.22084	-0.21809	-0.00602	$3.21	$4.46
1-m, 2-m, 6-m, 12-m	0.00186	1.21485	-0.13541	-0.14322	0.06191	$2.62	$4.62

The results show that using a single futures contract provides a reasonable replication, but performance diminishes as the maturity of the contract increases (out-of-sample RMSE rises from $5.15 to $17.95). Adding a second futures contract generally improves the out-of-sample RMSE, with the 2-month and 6-month combination performing best at $4.63. The best-performing portfolio overall is the three-contract combination of 2-month, 6-month, and 12-month futures, which achieved an out-of-sample RMSE of $4.46. Notably, several of these futures portfolios outperform the GLD ETF, which has an out-of-sample RMSE of $6.00 over the same period.

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Table 6: Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

Portfolio	w_0	w_{short}	w_{long}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
1-m	-0.01071	1.01071		\$6.63	\$3.34
2-m	-0.04835	1.04835		\$12.22	\$6.12
12-m	-0.06842	1.06842		\$15.11	\$5.71
1-m, 2-m	-0.00088	1.27315	-0.27227	\$6.27	\$2.79
1-m, 6-m	-0.00171	1.23899	-0.23729	\$6.27	\$2.98
1-m, 12-m	-0.00021	1.19336	-0.19315	\$6.29	\$3.00
2-m, 6-m	-0.04079	5.06602	-4.02523	\$10.27	\$9.37

Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	**Leverage (β)**	**Slope**	**Intercept**	**t-stat**	**p-value**	R^2	**RMSE**
GLD	1	1.0006	-8.89e-06	0.2595	0.7952	0.9714	0.00185
UGL	2	2.0209	-1.95e-04	2.7773	0.0055	0.9644	0.00349
GLL	-2	-2.0143	1.89e-04	-1.9231	0.0546	0.9650	0.00345
DGLD	-2	-1.9756	-8.49e-05	0.4684	0.6395	0.4066	0.02098
UGLDF	3	3.0063	4.59e-04	0.0582	0.9536	0.2552	0.04463

The regression results indicate that GLD, UGL, and GLL are effective at tracking their underlying benchmarks, with high R^2 values and slopes very close to their target betas. The p-value for GLD (0.7952) shows its slope is not statistically different from 1. However, the p-values for UGL (0.0055) and GLL (0.0546) suggest their slopes are statistically different from their target betas of 2 and -2, respectively. DGLD and UGLDF perform poorly, with very low R^2 values (0.4066 and 0.2552), indicating they do not reliably track their leveraged benchmarks.

Table 8: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	**Leverage (β)**	**Slope**	**Intercept**	**t-stat**	**p-value**	R^2	**RMSE**
GLD	1	1.00540	-1.6427×10^{-6}	1.42692	0.15382	0.98060	0.00163
UGL	2	2.00572	-1.3107×10^{-4}	0.73319	0.46357	0.97930	0.00337
GLL	-2	-2.00556	-1.0550×10^{-4}	0.67089	0.50240	0.97673	0.00358
UGLD	3	2.99358	-1.9860×10^{-4}	0.45211	0.65133	0.98484	0.00421
DGLD	-3	-2.97528	-1.6912×10^{-5}	0.97442	0.33019	0.95256	0.00752

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Static replication performs poorly when attempting to track leveraged benchmarks. While in-sample RMSEs are already significantly higher than for the spot benchmark, the out-of-sample RMSEs are extremely large (e.g., $322.25 for a single 1-month future replicating a 2x benchmark, and $1012.21 for a 3x benchmark). This demonstrates that a static portfolio of futures cannot capture the path-dependent nature and daily compounding effects inherent in leveraged ETFs, leading to large tracking errors over time.

The t-statistics far from zero for UGL and GLL indicate that the slopes are statistically significantly different from the specified leverage ratios.

Table 9: New Data: Weights and in/out of sample RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	**Portfolio**	w_0	w_1	w_2	w_6	w_{12}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
2x Leveraged	1-m	-1.60142	2.60142				\$53.41	\$322.25
	2-m	-1.59131		2.59131			\$50.95	\$324.58
	6-m	-1.55475			2.55475		\$43.04	\$341.84
	12-m	-1.47500				2.47500	\$44.85	\$345.55
	1-m, 2-m	-1.55169	-9.49762	12.04931			\$46.73	\$333.20
	1-m, 12-m	-1.51431	0.92308			1.59123	\$41.51	\$335.23
3x Leveraged	1-m	-3.96148	4.96148				\$175.92	\$1012.21
	2-m	-3.94307		4.94307			\$171.38	\$1015.74
	6-m	-3.87794			4.87794		\$153.52	\$1040.28
	12-m	-3.71788				4.71788	\$143.67	\$1034.43
	1-m, 2-m	-3.80855	-32.24870	37.05726			\$165.29	\$1085.27
	1-m, 12-m	-3.70334	-0.34149			5.04483	\$143.53	\$1037.21

Table 10: Leung & Ward (2015) Data: RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	**Portfolio**	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
+2x (UGL)	1-m	\$153.20	\$41.49
	2-m	\$140.41	\$49.57
	6-m	\$138.63	\$47.46
	12-m	\$134.51	\$48.30
+3x (UGLD)	1-m	\$555.49	\$111.51
	2-m	\$529.91	\$125.99
	6-m	\$526.58	\$122.33
	12-m	\$518.97	\$123.83